Saturating Interactions and the Stability of the Synchronous Equilibrium in the Five-Element Model of Traditional Chinese Medicine

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Abstract

Continuous-time models of the five-element (Wu-Xing) network of Traditional Chinese Medicine have almost without exception represented the inter-organ interactions by bilinear terms, following the Lotka–Volterra formalism of population dynamics. We show that this choice determines the qualitative dynamics: under bilinear coupling the fully synchronous equilibrium, which is the natural dynamical representation of health, is generically unstable. We propose an alternative in which the generating and over-acting interactions are saturating, each term being derived from a named classical relation, and analyse the resulting system. The positive orthant is forward invariant and a compact box is globally attracting for all parameter values, so that positivity and boundedness hold without restriction. The system is equivariant under the cyclic group $\mathbb{Z}_5$, whence the Jacobian at a synchronous equilibrium is circulant and its spectrum is obtained in closed form by discrete Fourier transform. This yields a necessary and sufficient condition for asymptotic stability consisting of three inequalities, whose coefficients are the golden ratio and its reciprocal, and which distinguishes the synchronous isotypic component from the symmetry-breaking ones. Numerical evaluation of the criterion indicates that stability is lost exclusively through symmetry-breaking modes, by a Hopf bifurcation whose critical eigenspace supports discrete rotating waves. In the complementary parameter regime the system is bistable, the synchronous equilibrium being annihilated in a saddle-node bifurcation whose locus is obtained explicitly, so that the loss of stability is discontinuous and exhibits hysteresis. An extension incorporating a demand term and a counter-acting term reproduces the classical pathological relations 母病及子, 子盗母气, 相乘 and 相侮, together with the classical ordering of the two abnormal control relations.

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1 Introduction

Traditional Chinese Medicine (TCM) represents the body as a network of interacting organ systems rather than as a collection of independent organs [Kap00; TLM08]. The five-element, or Wu-Xing, diagram places five such systems on a cycle of generation (Sheng, 生) and a pentagram of control (Ke, 克). Diagnostic and therapeutic principles are formulated directly in terms of these two relations: the precept 虚则补其母, that a deficient child is treated by tonifying its mother, refers to a single edge of the generating cycle, and the abnormal relations 相乘 and 相侮 refer to single edges of the control cycle [00]. Porkert observed that Chinese medicine is not less logical than Western medicine but less analytical [Por90]; the present work is concerned with the consequences of formulating these relations as a dynamical system.

Existing quantitative treatments are of two kinds. Discrete models assign to each organ finitely many states, either active and inactive in a synchronous Boolean network [BSL+98] or excess, normal and deficiency [Che02], or else reason algebraically about the relations without integration [Men; Zha11; KSK+14; FNW03]. The asymptotic behaviour of such models is necessarily a fixed point or a cycle, so questions of asymptotic stability in the sense considered here do not arise. Continuous models assign to each organ a real-valued activity governed by an ordinary differential equation. A small number employ linear interaction terms [CZ08; ZZ06]; the majority employ bilinear terms of the form $\beta x_i x_j$, by analogy with Lotka–Volterra population dynamics [Wan08; DW08; Che08; Yu05; Jia06; Li87; ZH98; Mur02]. Models within this family differ principally in the number of influences retained. Lin and Han [LH20] enumerate six and construct a symmetric five-dimensional system containing all of them; Nian and Li [NL25] have recently examined bifurcations in a related system. The asymptotic behaviours reported include equilibria [Wan08; DW08; Che08; Yu05], periodic orbits [Jia06; Li87; ZH98] and chaotic attractors [Jia06; Li87].

The bilinear convention has two consequences that motivate the present work. First, in the terminology of population ecology a bilinear coupling is a Holling type I functional response [Hol59], whose rate is unbounded in the interacting quantities. Second, it reverses the direction of one of the classical relations: the classical sources state that a deficient child draws more heavily upon its mother, whereas a term of the form $c\,x_i x_{\sigma(i)}$ tends to zero as the child is depleted. Both features are corrected by a saturating, or type II, response. We establish that the substitution is not merely cosmetic: under bilinear coupling the synchronous equilibrium is generically unstable, so that the model admits no representation of sustained health, whereas under saturating coupling it is asymptotically stable on a substantial region of parameter space.

The analysis is facilitated by the symmetry of the network. Since the five organs obey identical dynamical laws and the interaction structure is invariant under rotation by one position, the system is equivariant under the cyclic group $\mathbb{Z}_5$, and the Jacobian evaluated at a synchronous equilibrium is circulant. Its spectrum is therefore obtainable in closed form, and the isotypic decomposition separates the synchronous perturbations, under which all organs vary in unison, from the symmetry-breaking perturbations, under which they diverge. The latter correspond to the differentiated patterns of excess and deficiency identified in TCM diagnosis.

The paper is organised as follows. Section 2 formulates the model and derives each term from a named classical relation. Section 3 establishes forward invariance of the positive orthant, the existence of a globally attracting compact region, and the classification of synchronous equilibria. Section 4 derives the circulant structure of the Jacobian and the resulting stability criterion. Section 5 identifies the codimension-one bifurcations by which the synchronous equilibrium is destabilised or annihilated. Section 6 reports numerical simulations, including the reproduction of the classical pathological relations by an extended model. Section 7 interprets the classical treatment principles in terms of network motifs, and Section 8 discusses the scope and limitations of the results. Appendix A records corrections to Lyapunov spectra previously reported for the bilinear

model, and Appendix B documents the numerical methods and their validation.

2 Model formulation

2.1 The interaction network

The five organ systems are indexed according to the standard cycle:

Index	Element	Organ
x_1	Wood (木)	Liver (肝)
x_2	Fire (火)	Heart (心)
x_3	Earth (土)	Spleen (脾)
x_4	Metal (金)	Lung (肺)
x_5	Water (水)	Kidney (肾)

Let $x_i(t) \geq 0$ denote the functional activity, or qi, of organ i at time t; indices are understood modulo five throughout. The generating cycle is $x_1 \to x_2 \to x_3 \to x_4 \to x_5 \to x_1$, in which each organ is termed the *mother* of its successor and the successor its *child*. The control cycle follows the pentagram $x_1 \dashv x_3 \dashv x_5 \dashv x_2 \dashv x_4 \dashv x_1$. Denoting by $\sigma(i)$ the mother of organ i, by $\kappa(i)$ the organ it controls, and by $\rho(i)$ the organ controlling it,

$$\sigma(i) = i - 1, \qquad \sigma^{-1}(i) = i + 1, \qquad \kappa(i) = i + 2, \qquad \rho(i) = i - 2. \tag{1}$$

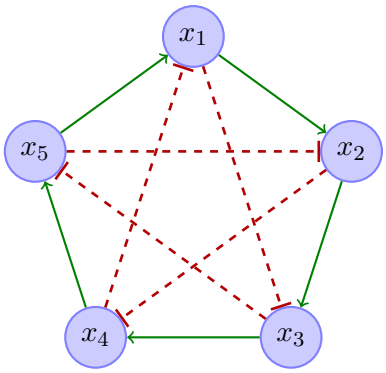


Figure 1: The Wu-Xing interaction network. Solid arrows denote the generating (Sheng) cycle; dashed bars denote the controlling (Ke) pentagram. Each organ possesses exactly one mother, one child, one controller and one target, and the configuration is invariant under rotation by one position.

2.2 The classical relations

The relations to be represented are those of the classical literature [00], following the exposition of [Kap00; Por90]. Along the generating cycle, the mother nourishes the child and a deficient child draws more heavily upon her; deficiency in the mother induces deficiency in the child (母病及子) and excess induces excess; an excessive child overdraws from the mother (子盗母气); and sustained weakness in the child eventually depletes the mother. Along the control cycle, ordinary restraint is physiological and serves to prevent unbounded growth of the target; where the controller is excessive or the target deficient, the target is severely weakened, the relation termed 相乘 and classified as

贼邪, the "thief evil"; and conversely, where the target is excessive or the controller deficient, the controller is mildly weakened, the relation termed 相侮 and classified as 微邪, the "mild evil". The explicit ordering of these two abnormal relations constrains the corresponding coefficients and is used in Section 6.3.

2.3 Governing equations

We consider the system

$$\dot{x}_i \;=\; \underbrace{a x_i - k x_i^2}_{\text{self-regulation}} \;+\; \underbrace{\frac{c\, x_{\sigma(i)}}{1 + x_i}}_{\text{generation}} \;-\; \underbrace{\frac{g\, x_i}{1 + x_{\kappa(i)}}}_{\text{over-acting}} \;-\; \underbrace{h\, x_i x_{\rho(i)}}_{\text{restraint}}, \qquad i \in \mathbb{Z}_5, \tag{2}$$

with $a, k, c, g, h > 0$. The four terms are derived as follows.

The self-regulation term is logistic, comprising production proportional to current activity together with a quadratic limitation, and possesses intrinsic carrying capacity a/k.

In the generation term, the numerator renders nourishment proportional to the mother's activity, so that deficiency and excess propagate from mother to child in accordance with the classical relations. The denominator represents demand: the factor $(1 + x_i)^{-1}$ approaches unity as the child is depleted and decreases as the child becomes replete. This is the relation that a bilinear term cannot represent, since the product $c\, x_i x_{\sigma(i)}$ vanishes precisely in the limit of greatest need.

The over-acting term represents 相乘. The factor $(1 + x_{\kappa(i)})^{-1}$ approaches unity when the controlled organ is deficient, so that the demand upon x_i is then maximal, and decreases as the controlled organ approaches its normal level. Deficiency in the target thus amplifies the cost borne by the controller.

The restraint term is retained in bilinear form. Ordinary suppression requires both parties to be present, and a product is therefore the appropriate representation. In the terminology of population ecology, the model replaces type I functional responses by type II responses on the generating and over-acting edges while retaining a type I response on the restraint edge [Hol59; Mur02].

System (2) omits the counter-acting relation 相侮 and the consumption borne by the mother. Both are reinstated in Section 6.3, where they are shown to be required for the reproduction of the full set of classical pathologies. The reduced system (2) is analysed first, being the smallest model exhibiting the phenomena under consideration; the analysis extends without modification.

Remark 2.1 (Scaling). The substitutions $t \mapsto at$ and $x \mapsto (a/k)x$ reduce the five parameters to three, at the cost of introducing an explicit scale into the saturation denominators, which become $1 + (a/k)x_i$. Since the classical relations are expressed in terms of an organ being deficient or excessive relative to a fixed physiological norm, the dimensional form is retained, the saturation threshold furnishing the natural unit of activity and each coefficient retaining its interpretation. The values $a = 2$, $k = 0.5$, $c = 1$, $g = 0.5$ and $h = 0.3$ are used throughout for illustration.

3 Positivity, boundedness and equilibria

3.1 Forward invariance and dissipativity

Since x_i denotes a physiological activity, the model is admissible only if solutions remain nonnegative and bounded. For (2) both properties hold without restriction on the parameters.

Proposition 3.1 (Invariant region)**.** *The positive orthant is forward invariant under* (2)*, and the compact set* $[0, X_{\max}]^5$ *with*

$$X_{\max} = \frac{a + \sqrt{a^2 + 4kc}}{2k} \tag{3}$$

is positively invariant and globally attracting. In particular, no solution ceases to exist in finite time, for any admissible parameter values.

Proof. Every term of (2) contains a factor x_i with the exception of the generation term, which is nonnegative. Hence $\dot{x}_i \geq 0$ whenever $x_i = 0$, and the boundary of the positive orthant cannot be crossed from within.

For the second assertion, set $X(t) = \max_i x_i(t)$ and let i attain the maximum. The generation term satisfies $c\,x_{\sigma(i)}/(1 + x_i) \leq cX/(1 + X) < c$, while the over-acting and restraint terms are nonpositive. Therefore

$$\dot{X} \;\leq\; aX - kX^2 + c,$$

and the right-hand side is negative whenever $kX^2 - aX - c > 0$, that is whenever $X > X_{\max}$ as given by (3). □

The corresponding property fails for the bilinear model. Restricted to the synchronous subspace, that model reduces to $\dot{x} = ax - Mx^2$ with $M = b + d + g + h - c$, whose solutions cease to exist in finite time whenever $M < 0$. Boundedness is thus a condition on the parameters rather than a structural property of the model, and the parameter values for which it fails are not excluded on physiological grounds.

3.2 Synchronous equilibria

Definition 3.2. An equilibrium is *synchronous* if $x_1 = \cdots = x_5 = x^*$. Synchronous equilibria with $x^* > 0$ constitute the model's representation of health; the origin represents complete loss of function.

Substituting $x_i = x^*$ into (2), every neighbour of every organ is equal to x^* and the five equations reduce to one. Dividing by $x^* > 0$ and clearing the denominator yields

$$(k + h)\,(x^*)^2 \;+\; \big[(k + h) - a\big]\,x^* \;-\; (a + c - g) \;=\; 0. \tag{4}$$

The leading coefficient being positive, the sign of the constant term $-(a + c - g)$ determines the structure of the root set.

Proposition 3.3. *If $a + c > g$, equation (4) possesses exactly one positive root,*

$$x^* = \frac{a - (k + h) + \sqrt{\big[a - (k + h)\big]^2 + 4(k + h)(a + c - g)}}{2(k + h)}, \tag{5}$$

and system (2) accordingly possesses a unique synchronous equilibrium in the interior of the positive orthant.

Proof. The product of the roots of (4) equals $-(a + c - g)/(k + h)$, which is negative under the stated hypothesis. Exactly one root is therefore positive, and it is given by (5). □

The condition $a + c > g$ requires that self-production together with nourishment received exceed the cost of nourishing the child. At the illustrative parameter values it yields $x^* = \frac{3}{4} + \frac{\sqrt{59}}{4} \approx 2.6703$. The complementary regime is treated in Section 5.2, where it is shown to admit two positive synchronous equilibria rather than none.

4 Linear stability of the synchronous equilibrium

4.1 Equivariance and the circulant structure of the Jacobian

Let γ denote the cyclic shift $(\gamma x)_i = x_{i-1}$ on $\mathbb{R}^5$. The maps (1) commute with γ, and the five organs obey identical laws; consequently the vector field F of (2) satisfies $F(\gamma x) = \gamma F(x)$, so that (2) is equivariant under the cyclic group $\mathbb{Z}_5 = \langle\gamma\rangle$. Equivariance is inherited by the linearisation at any γ-invariant point, and every synchronous equilibrium is such a point.

Lemma 4.1. *Let J denote the Jacobian of (2) at a synchronous equilibrium x^*, and set $s = 1 + x^*$. Then*

$$J_{ii} = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \qquad \frac{\partial \dot{x}_i}{\partial x_{\sigma(i)}} = \frac{c}{s}, \qquad \frac{\partial \dot{x}_i}{\partial x_{\kappa(i)}} = \frac{gx^*}{s^2}, \qquad \frac{\partial \dot{x}_i}{\partial x_{\rho(i)}} = -hx^*,$$

all remaining entries being zero, and J is circulant.

Proof. The partial derivatives are obtained by differentiating (2) and evaluating at $x_i = x^*$ for all i; the two saturating terms require the quotient rule. The three nonzero off-diagonal entries occupy the positions $\sigma(i) = i - 1$, $\kappa(i) = i + 2$ and $\rho(i) = i - 2$, which stand at offsets independent of i. Each row is therefore the preceding row advanced by one position, which is the definition of a circulant matrix. □

4.2 Closed-form spectrum and stability criterion

Circulant matrices are diagonalised by the discrete Fourier basis [Dav79]. Writing $\omega = e^{2\pi i/5}$, the eigenvalues of J are the values of the symbol of its first row at the fifth roots of unity,

$$\lambda_j = J_{ii} + \frac{gx^*}{s^2}\,\omega^{2j} - hx^*\,\omega^{3j} + \frac{c}{s}\,\omega^{4j}, \qquad j = 0, \dots, 4. \tag{6}$$

Since $\omega^3 = \overline{\omega^2}$ and $\omega^4 = \overline{\omega}$, only two distinct cosines occur in the real parts, and both admit expression in terms of the golden ratio $\varphi = (1 + \sqrt{5})/2$:

$$\cos\frac{2\pi}{5} = \frac{1}{2\varphi}, \qquad \cos\frac{4\pi}{5} = -\frac{\varphi}{2}.$$

Theorem 4.2. *Let $x^* > 0$ be a synchronous equilibrium of (2), let $s = 1 + x^*$, and define*

$$D = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \qquad T = x^*\Big(\frac{g}{s^2} - h\Big), \qquad M = \frac{c}{s}.$$

Then J possesses exactly three distinct eigenvalue real parts,

$$\operatorname{Re}\lambda_0 = D + T + M, \qquad \operatorname{Re}\lambda_{1,4} = D - \tfrac{\varphi}{2}T + \tfrac{1}{2\varphi}M, \qquad \operatorname{Re}\lambda_{2,3} = D + \tfrac{1}{2\varphi}T - \tfrac{\varphi}{2}M, \tag{7}$$

and x^* is asymptotically stable if and only if all three are negative.

Proof. Taking real parts in (6) gives $\operatorname{Re}\lambda_j = D + T\cos(4\pi j/5) + M\cos(8\pi j/5)$, the contributions of the target and controller combining because $\cos(6\pi j/5) = \cos(4\pi j/5)$ for every j. Evaluation of the cosines at $j = 0, \dots, 4$ yields only the values 1, $1/(2\varphi)$ and $-\varphi/2$, arranged as stated. The equivalence of asymptotic stability with negativity of all eigenvalue real parts at a hyperbolic equilibrium is standard. □

At the illustrative parameter values, (7) gives $\operatorname{Re}\lambda_0 \approx -2.2353$, $\operatorname{Re}\lambda_{1,4} \approx -1.1537$ and $\operatorname{Re}\lambda_{2,3} \approx -2.2432$, so that the synchronous equilibrium is asymptotically stable. For the bilinear model at the parameter values of [LH20], the corresponding computation yields the pair $0.206 \pm 0.634i$, of positive real part, and the synchronous equilibrium is unstable. Figure 2 displays the two spectra.

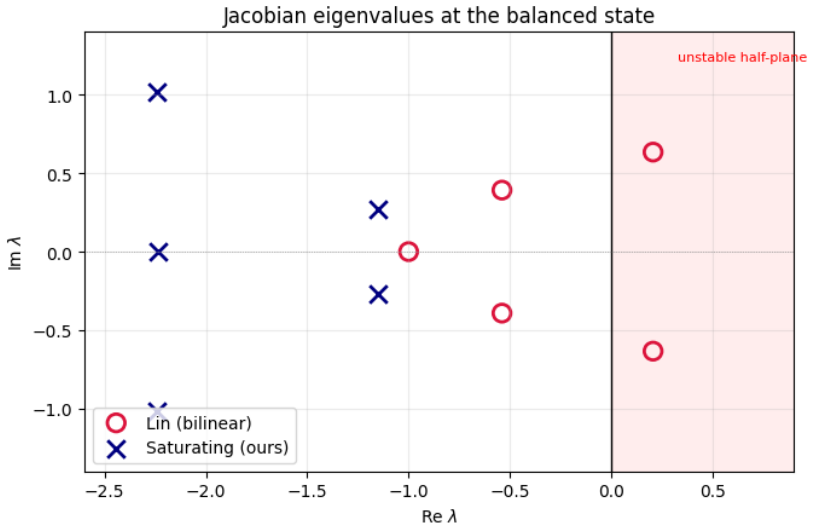


Figure 2: Spectra of the Jacobian at the synchronous equilibrium of each model. The shaded half-plane is the region of positive real part. Two of the five bilinear eigenvalues lie within it; all five eigenvalues of the saturating model lie strictly to the left of the imaginary axis.

4.3 Isotypic decomposition

The three expressions in (7) correspond to the isotypic components of the $\mathbb{Z}_5$ action. The mode $j = 0$ has eigenvector $(1, 1, 1, 1, 1)$ and spans the synchronous subspace; a perturbation in this direction alters the activity of all five organs equally, leaving them mutually balanced. The modes $j = 1, \dots, 4$, which occur in complex-conjugate pairs, span the complementary symmetry-breaking subspace; perturbations in these directions differentiate the organs, producing a pattern of excess and deficiency across the network. The latter correspond to the differentiated patterns identified in TCM diagnosis, and which of the three conditions in (7) is violated first therefore determines the character of the resulting disorder.

The three conditions weight T and M differently. The synchronous mode superposes both at unit weight, whereas the symmetry-breaking modes oppose them with weights $-\varphi/2$ and $1/(2\varphi)$. An increase in the restraint coefficient h renders T more negative, which relaxes the condition for $j = 1, 4$ and tightens that for $j = 2, 3$; the resulting competition determines the stability boundary.

5 Bifurcation analysis

The synchronous equilibrium may cease to be an asymptotically stable representation of health in three distinct ways, each admitting an explicit condition. Two concern its existence and follow from the quadratic (4); the third concerns its stability and follows from Theorem 4.2.

5.1 Transcritical bifurcation at the origin

The origin is an equilibrium of (2) for all parameter values, and its stability admits a closed-form characterisation.

Proposition 5.1. *The Jacobian of (2) at the origin is $J(0) = (a - g)I + cP$, where P is the permutation matrix of the map σ. Its eigenvalues are $\lambda_j = (a - g) + c\,\omega^j$ for $j = 0, \dots, 4$, whence the spectral abscissa is $a + c - g$ and the origin is asymptotically stable if and only if $a + c < g$.*

Proof. At $x = 0$ one has $s = 1$, so that the diagonal entries of Lemma 4.1 reduce to $a - g$, and the entries associated with the target and controller vanish, each containing a factor x^*. Only the entry c in the position $\sigma(i)$ survives, whence $J(0) = (a - g)I + cP$. The eigenvalues of the cyclic permutation P are the fifth roots of unity, and $\operatorname{Re}\omega^j = \cos(2\pi j/5)$ attains its maximum at $j = 0$. □

The condition of Proposition 5.1 is the negation of that of Proposition 3.3. The two results are therefore complementary: for $a + c > g$ the origin is unstable and a unique positive synchronous equilibrium exists, while for $a + c < g$ the origin is asymptotically stable. At $a + c = g$ the positive branch passes through the origin and the two equilibria exchange stability, which is a transcritical bifurcation.

5.2 Saddle-node bifurcation and bistability

For $a + c < g$ the constant term of (4) is positive and its two roots are of like sign. Their sum is $[a - (k + h)]/(k + h)$, which is positive when $a > k + h$. Failure of the condition $a + c > g$ therefore does not imply that the origin is the only nonnegative equilibrium; for $a > k + h$ there may exist two positive synchronous equilibria.

Proposition 5.2. *Let $a > k + h$. The discriminant of (4) vanishes at*

$$g = g_{\mathrm{f}} := a + c + \frac{\big[a - (k + h)\big]^2}{4(k + h)}, \tag{8}$$

where the two positive roots coalesce at $x^*_{\mathrm{f}} = [a - (k + h)]/[2(k + h)]$. For $a + c < g < g_{\mathrm{f}}$ there exist two positive synchronous equilibria and for $g > g_{\mathrm{f}}$ none, so that the system undergoes a saddle-node bifurcation at $g = g_{\mathrm{f}}$.

Proof. The discriminant of (4) is $[a - (k + h)]^2 + 4(k + h)(a + c - g)$. Setting it equal to zero and solving for g yields (8), and the repeated root is $-B/(2A)$ with $A = k + h$ and $B = (k + h) - a$. □

Evaluation of the spectrum on each branch yields the configuration characteristic of a saddle-node bifurcation: the upper branch is asymptotically stable, the lower is of saddle type, and by Proposition 5.1 the origin is asymptotically stable throughout the interval. The system is accordingly bistable for $a + c < g < g_{\mathrm{f}}$, the lower branch constituting the separatrix between the two basins of attraction. Figure 3 displays the bifurcation diagram together with two solutions initiated on either side of the separatrix.

Three consequences follow. Within the bistable interval the asymptotic state is determined by the initial condition rather than by the parameters, so that systems with identical parameter values may converge either to the positive equilibrium or to the origin according to their position relative to the separatrix. The positive branch is not attenuated continuously but annihilated at a finite value of x^*, so that a slow variation of g through g_{f} produces a discontinuous loss of function. Since the origin remains asymptotically stable throughout the interval, reversal of that variation does not restore the system to the positive branch; recovery requires either reduction of g below the transcritical value or a perturbation returning the state to the basin of the positive equilibrium. The transition therefore exhibits hysteresis.

5.3 Symmetry-breaking Hopf bifurcation

Where the positive synchronous equilibrium exists it may nonetheless lose stability through violation of Theorem 4.2, and the isotypic component in which the violation occurs determines the character of the bifurcation.

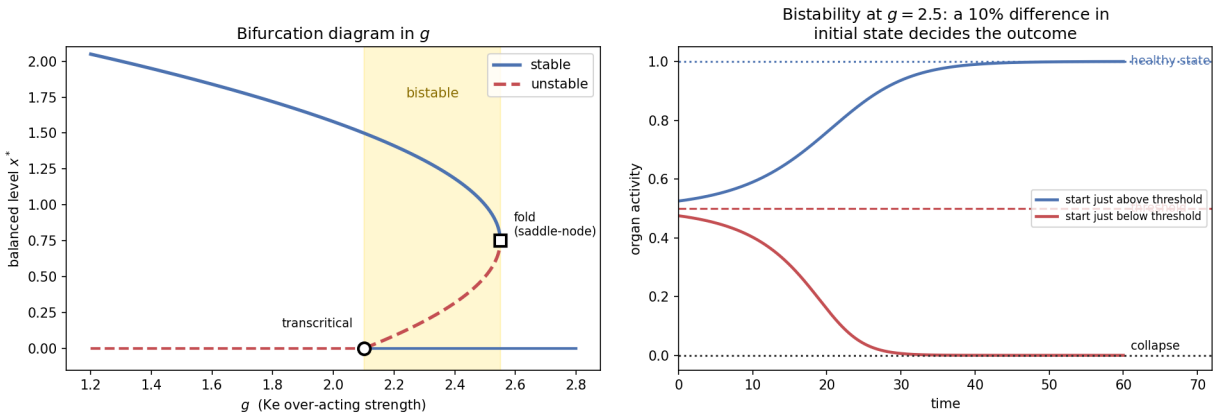


Figure 3: Left: bifurcation diagram in the over-acting coefficient g for $a = 2$, $k = 0.5$, $c = 0.1$ and $h = 0.3$. Solid curves denote asymptotically stable equilibria and dashed curves unstable ones. The upper branch is annihilated at $g_{\mathrm{f}} = 2.55$ at the finite value $x^* = 0.75$, and the shaded interval is bistable. Right: two solutions within that interval, initiated ten percent above and ten percent below the separatrix, converging respectively to the positive equilibrium and to the origin.

Evaluating (7) on a 200×200 grid in the generation coefficient c and the restraint coefficient h, the synchronous mode $j = 0$ is critical in three percent of grid cells, and at each of the 680 cells lying within 0.02 of the stability boundary the critical eigenvalues are the pair $j = 1, 4$ (Figure 4). Within the region examined, the synchronous equilibrium therefore does not lose stability through a synchronous perturbation. A synchronous instability would correspond to uniform depletion or uniform over-activity with the organs remaining mutually balanced; the computation indicates instead that stability is lost through a differentiated pattern of excess and deficiency, which is the form in which TCM diagnosis is expressed. This is a consequence of the symmetry of the network rather than an assumption of the model.

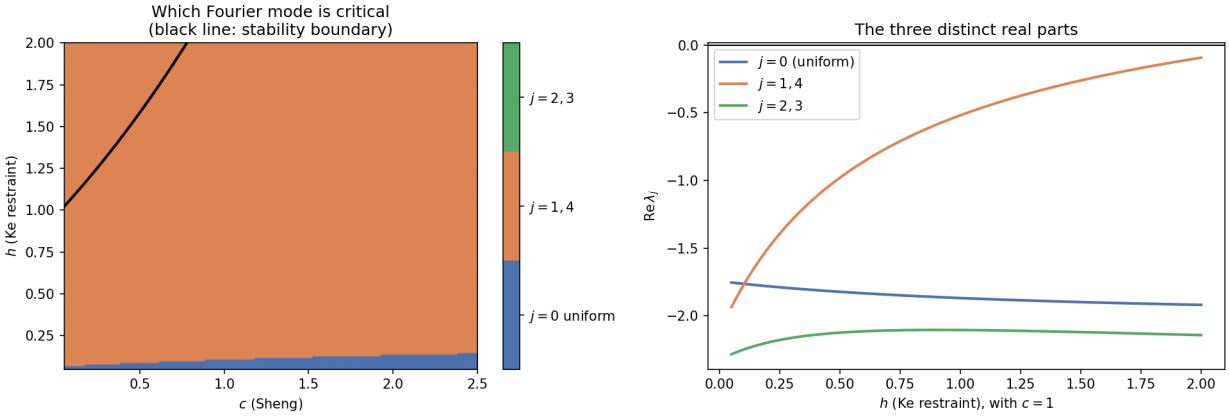


Figure 4: Left: the critical isotypic component across the (c, h) plane, with the stability boundary superimposed. Right: the three eigenvalue real parts of (7) along the section $c = 1$, showing the pair $j = 1, 4$ crossing the imaginary axis while the synchronous mode remains strictly negative.

Since $j = 1, 4$ is a complex-conjugate pair, the crossing is oscillatory provided the imaginary part is nonzero at criticality. Bisection in h at $a = 1.5$, $k = 0.65$, $c = 1$ and $g = 0.5$ locates the

boundary at $h_c = 2.3718$, where $x^* = 0.5998$ and the spectrum is

$$\pm 0.310613\, i, \qquad -1.929714, \qquad -2.158417 \pm 1.831876\, i,$$

a single conjugate pair on the imaginary axis with the remaining eigenvalues of negative real part. The transversality and nondegeneracy conditions being satisfied generically, the transition is a Hopf bifurcation, and the period of the emergent oscillation at onset is $2\pi/|\operatorname{Im}\lambda| = 20.23$. Repetition of the bisection at other values of c yields the same classification throughout, the imaginary part decreasing as c increases (Table 1).

c	h_c	critical pair
0.5	1.5843	$\pm 0.4113\, i$
1.0	2.3718	$\pm 0.3106\, i$
1.5	3.3475	$\pm 0.1976\, i$
2.0	4.0000	$\pm 0.0790\, i$

Table 1: The stability boundary at several values of the generation coefficient. In each case a single complex-conjugate pair crosses the imaginary axis with nonzero imaginary part.

The critical eigenspace being a symmetry-breaking isotypic component of a $\mathbb{Z}_5$-equivariant system, the equivariant Hopf theorem predicts branches of discrete rotating waves, in which the organs oscillate with a constant phase lag of one fifth of the period [GSS88; GS02]. Periodic solutions of this form have been reported for bilinear models [LH20], which suggests that the two formulations lose stability into solutions of the same isotropy type notwithstanding their different couplings. The criticality of the bifurcation is determined by the first Lyapunov coefficient, which has not been computed here; the distinction is material, since a subcritical Hopf bifurcation would render the onset of oscillation discontinuous and hysteretic in the manner of the saddle-node bifurcation of Section 5.2.

6 Numerical simulations

6.1 Response to perturbation of a single organ

Theorem 4.2 concerns infinitesimal perturbations. To assess its consequences for perturbations of finite magnitude, solutions were initiated at the synchronous equilibrium with a single organ displaced, and the deviation $\|x(t) - x^*\mathbf{1}\|_2$ recorded.

In the saturating model the deviation decays to zero following both a five percent displacement and a severe deficiency. In the bilinear model a five percent displacement is amplified by a factor of approximately fifty, and a severe deficiency drives several organs towards zero. Since the interaction structure is invariant under rotation, displacement of any organ must produce an identical response; this was verified, the responses agreeing across the five choices to within 10^{-14} (Figure 6). The agreement also constitutes a consistency check on the implementation of the maps (1), an error in which would break the equivariance.

6.2 Extent of the region of stability

Theorem 4.2 determines stability at a given point of parameter space but not the measure of the stable set, which depends upon the distribution assumed over the coefficients. Figure 7 evaluates the criterion across the plane of generation against restraint: the saturating model is stable throughout the region shown, the bilinear model on less than one fifth of it.

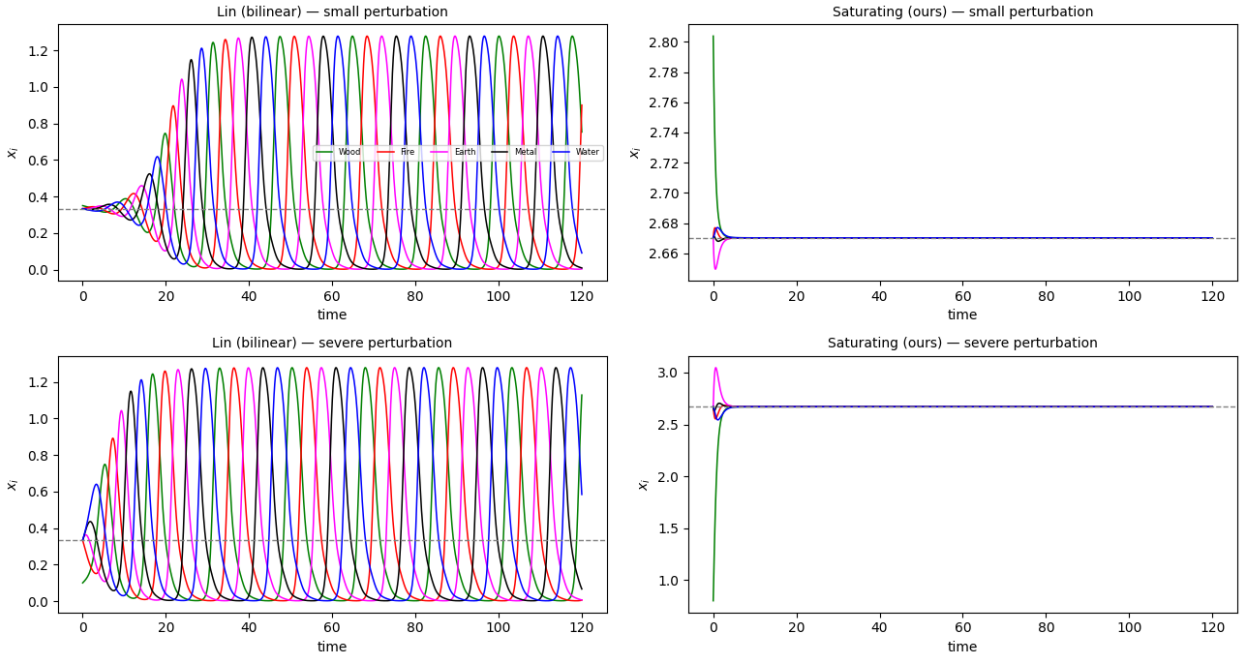


Figure 5: Response to displacement of a single organ by five percent (upper row) and to thirty percent of the equilibrium value (lower row). The dashed line marks the synchronous equilibrium.

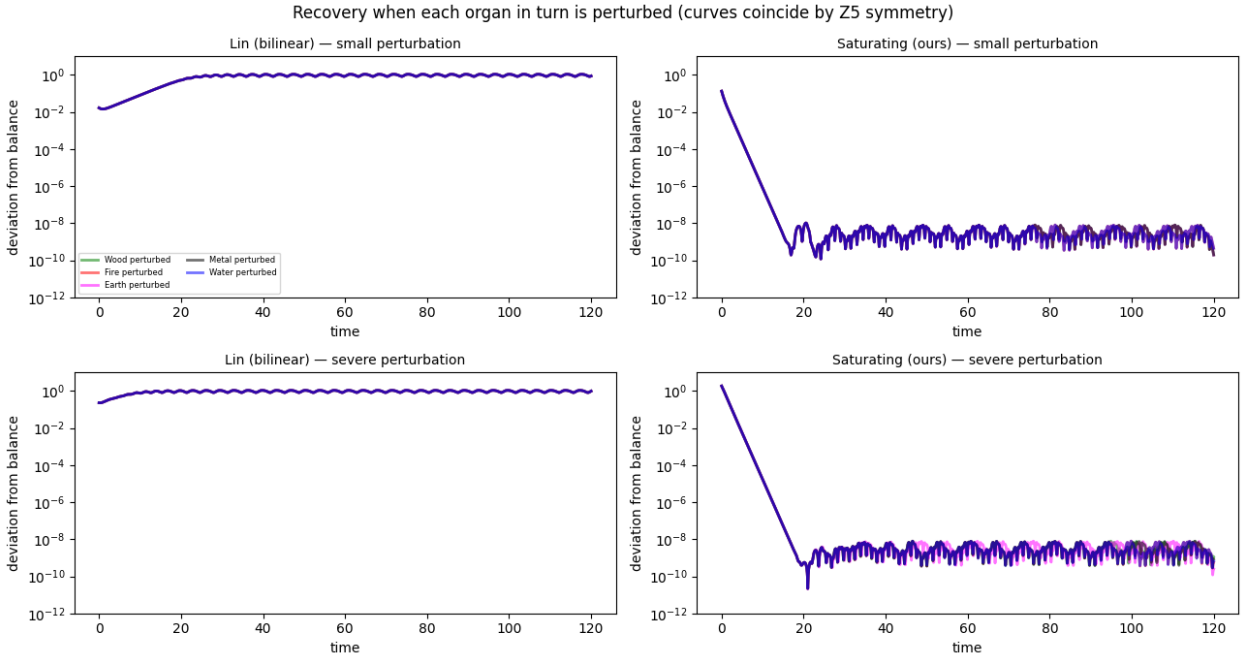


Figure 6: Deviation from the synchronous equilibrium, on a logarithmic scale, following displacement of each organ in turn. The five curves coincide, as equivariance requires. In the saturating model the deviation decays by eight orders of magnitude before reaching the tolerance of the integrator.

To establish that this is not an artefact of the particular section, coefficients were additionally sampled at random. No canonical distribution over them being available, three were used: inde-

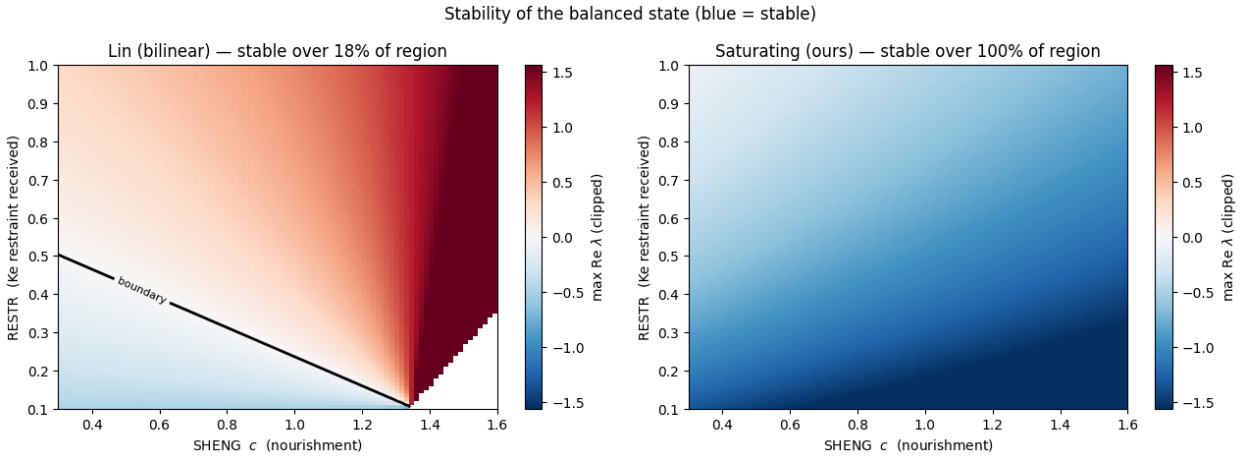


Figure 7: Spectral abscissa at the synchronous equilibrium across the (c, h) plane, the remaining coefficients being held at mid-range values. Blue denotes asymptotic stability. The pale region in the left panel is that in which the bilinear model possesses no positive synchronous equilibrium.

pendent uniform variates; variates conditioned on the ordering `SELF` > `SHENG` > `RESTR` > `TGT`, under which self-production exceeds generation, generation exceeds ordinary restraint, and ordinary restraint exceeds over-acting; and variates drawn from intervals assigned to each quantity in accordance with that ordering. Draws for which a model possesses no positive synchronous equilibrium are excluded from the corresponding count, each model being assessed only where it is well posed. The resulting proportions of stable draws are 22.8%, 47.1% and 23.3% for the bilinear model, against 94.1%, 91.9% and 99.0% for the saturating model, the Wilson intervals being disjoint in each case.

The imposition of the ordering approximately doubles the proportion for the bilinear model while slightly reducing that for the saturating model; the conditioned distribution is therefore the more favourable to the bilinear formulation, and the comparison under it the more conservative. It should further be noted that the two models do not possess the same number of coefficients, the bilinear model carrying an additional term representing the cost of nourishing the child. Removal of that term reduces its stable proportion from 22.1% to 12.0%, so that the asymmetry operates in favour of the bilinear model.

6.3 Reproduction of the classical pathological relations

The foregoing results are dynamical. We now assess whether the model reproduces the pathological relations enumerated in Section 2.2.

A chronic disorder is represented by a sustained modification of the self-production coefficient of a single organ, $a_j = \lambda a$, with $\lambda = 0.4$ representing deficiency and $\lambda = 1.6$ excess. The system is integrated to its new asymptotic state and the relative change in each organ recorded.

Applied to (2), this procedure reproduces 母病及子, 相乘 and 相侮, but not 子盗母气: rendering the child excessive increases the mother by 12.9%. The discrepancy has a structural origin. In the Wu-Xing topology the control target of Fire is Metal, and Metal is the controller of Wood, so that $\kappa(2) = 4 = \rho(1)$. An excess of Fire consequently suppresses Metal, and the suppression of Metal relieves Wood of restraint, so that Wood increases. The excess of the child thus reaches the mother along two paths of opposite sign, and the classical relation asserts that the direct path predominates; a model omitting that path necessarily obtains the opposite sign.

The construction of the omitted term requires care. A term of conservation type, $-g_s x_i/(1 +$

$x_{\sigma^{-1}(i)}$), representing the transfer received by the child, is decreasing in the child's activity and therefore cannot represent an excessive child consuming more. The relation 子盗母气 describes demand rather than conservation, and is represented by

$$-\frac{g_d\, x_{\sigma^{-1}(i)}\, x_i}{1+x_i},$$

in which the loss borne by the mother increases with the activity of the child.

Remark 6.1. The classical statements exhibit a tension at this point. A deficient child is said to draw more heavily upon its mother, while an excessive child is said to overdraw from her; the two assertions are opposed in direction and cannot proceed from a single term. We interpret them as describing distinct mechanisms, namely a physiological demand for nourishment, represented by the factor $(1+x_i)^{-1}$ in the generation term, and a pathological over-consumption, represented by the term above. The extended model contains both.

Incorporating the demand term together with the counter-acting term of Section 2.2 yields

$$\dot{x}_i = a_i x_i - k x_i^2 + \frac{c\, x_{\sigma(i)}}{1+x_i} - \frac{g_d\, x_{\sigma^{-1}(i)}\, x_i}{1+x_i} - \frac{g\, x_i}{1+x_{\kappa(i)}} - h\, x_i x_{\rho(i)} - \frac{b\, x_{\kappa(i)}}{1+\epsilon x_i}. \tag{9}$$

Both additional terms are nonpositive and both are defined through the maps (1). Proposition 3.1 and Lemma 4.1 therefore apply without modification: system (9) remains positive, dissipative and $\mathbb{Z}_5$-equivariant, and its Jacobian at a synchronous equilibrium remains circulant.

Relation	Modification	Organ specified	Predicted	Observed
母病及子	Wood deficient	Fire (child)	decrease	−19.3%
子盗母气	Fire excessive	Wood (mother)	decrease	−6.1%
相乘	Wood excessive	Earth (target)	marked decrease	−31.3%
相侮	Earth excessive	Wood (controller)	slight decrease	−17.7%

Table 2: Classical pathological relations reproduced by (9) at $g_d = 0.35$, $b = 0.10$ and $\epsilon = 0.5$.

All four relations are reproduced with the correct sign, and the classical ordering is respected, the effect of 相乘 (−31.3%) exceeding in magnitude that of 相侮 (−17.7%). A sweep of the two additional coefficients shows that all five criteria are satisfied simultaneously on 53% of the region examined (Figure 9), so that the agreement does not depend on a particular choice. The failures at the boundary of that region are informative: for b large relative to g_d the counter-acting effect exceeds the over-acting effect and the classical ordering is violated, while for g_d small the demand of the child does not overcome the indirect control path and 子盗母气 fails. The classical statements therefore constrain the coefficients from both directions, and in particular require the counter-acting coefficient to be small, which is the content of its classification as 微邪.

Figure 10 shows that the effect upon the organ specified by the classical relation is not in general the largest in the network, the control target of the modified organ being frequently displaced further. This is consistent with the classical emphasis upon the control cycle as the principal route of transmission, and with the corresponding therapeutic principle of protecting the organ next in that sequence rather than treating only the organ already affected.

7 Network motifs

[To be written. The argument to be developed: each classical therapeutic principle refers to only two or three organs—虚则补其母 and 实则泻其子 refer to a single generating edge, 相乘 and 相

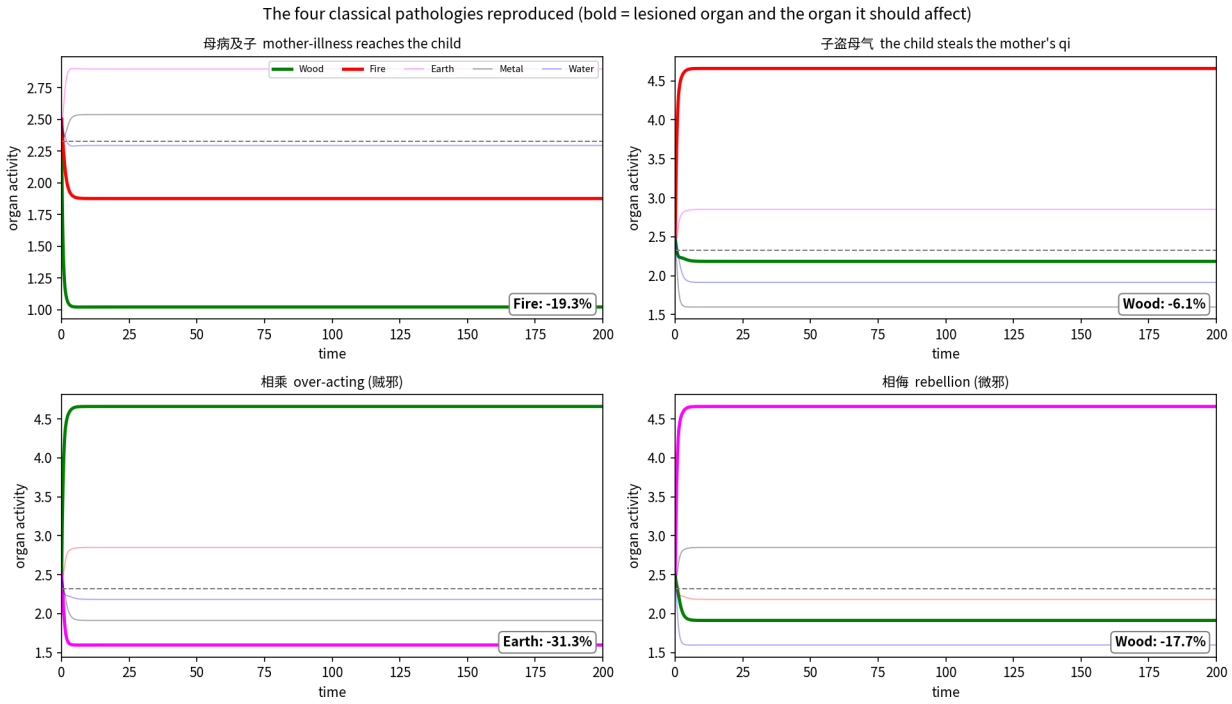


Figure 8: The four classical pathological relations. A single organ is modified at $t = 0$ and the system approaches a new asymptotic state; the dashed line marks the original synchronous equilibrium. The modified organ and the organ specified by the relation are drawn in bold.

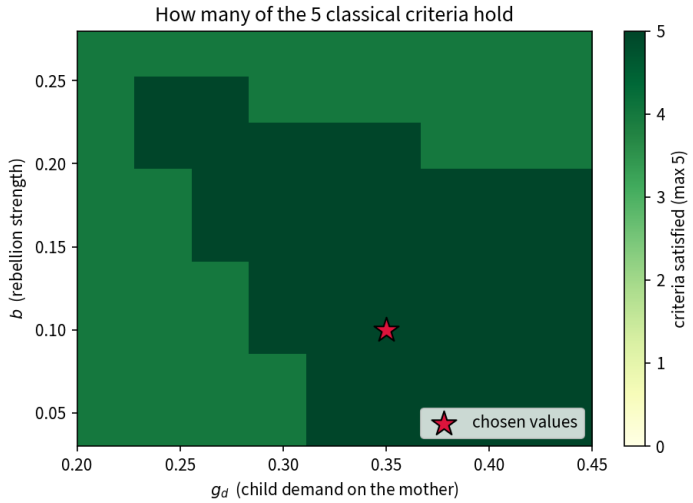


Figure 9: Number of the five classical criteria satisfied across the plane of the demand coefficient against the counter-acting coefficient. The marked point indicates the values of Table 2.

侮 to a single control edge, and the principle of strengthening the organ next to be affected refers to a three-node configuration—so that TCM constitutes an implicit motif theory of physiology in the sense of [Mil+02; She+02; Alo07; Alo06; MA03]. Reasoning of this local character is justified precisely when the coupling to the remainder of the network is weak or slow by comparison with the dynamics internal to the motif, which is a statement concerning separation of time scales

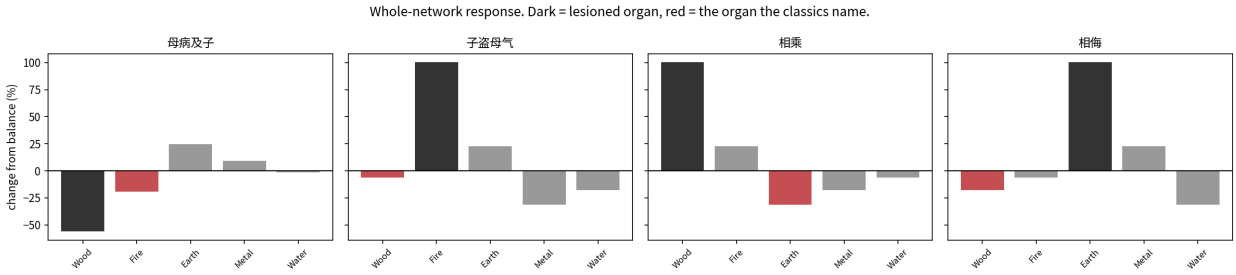


Figure 10: Response of the entire network to each modification. Dark bars indicate the modified organ and light bars the organ specified by the classical relation.

and is testable within the present model. The isotypic decomposition of Section 4.3 provides the connection between the local motifs and the global dynamics, each isotypic component being a delocalised pattern whose projection onto a given motif may be computed directly.]

8 Discussion

The results presented above indicate that the functional form assumed for the inter-organ interactions determines whether the five-element system admits a representation of sustained health. Under bilinear coupling, nourishment is proportional to the product of the activities of mother and child. This reverses the classical relation, a depleted child then receiving less rather than more, and leaves the interaction unbounded, so that it may exceed the quadratic self-limitation. The synchronous equilibrium is in consequence unstable, and outside the region in which the loss coefficients dominate the gain, solutions cease to exist in finite time. Substitution of a saturating response corrects both features: the factor $(1+x_i)^{-1}$ renders nourishment greatest when the child is most depleted, and bounds the interaction, so that the quadratic term ultimately dominates. The consequences are the invariant region of Proposition 3.1, positivity and dissipativity independent of the parameters, and asymptotic stability of the synchronous equilibrium on a substantial region of parameter space.

The circulant structure of the Jacobian is of interest beyond its computational convenience. The isotypic decomposition separates the synchronous component, in which the organs vary in unison, from the symmetry-breaking components, in which they are differentiated; the numerical evidence of Section 5.3 indicates that stability is lost exclusively through the latter. The component in which the violation first occurs therefore determines the character of the resulting disorder, and the $\mathbb{Z}_5$ symmetry constrains the admissible patterns, predicting in particular discrete rotating waves at the Hopf bifurcation.

Three limitations should be noted. First, the identification of health with a stable *synchronous* equilibrium is an assumption. The physiological system is not exactly symmetric, and health might alternatively be represented by a bounded oscillation; the results establish that the saturating model admits a stable synchronous equilibrium where the bilinear model does not, and not that an equilibrium is the only admissible representation. Second, Theorem 4.2 is a criterion for linear stability. Proposition 3.1 furnishes an absorbing region and Section 6 demonstrates recovery from perturbations of finite magnitude, but no Lyapunov function has been constructed and no claim of global asymptotic stability is made; indeed Section 5.2 establishes that global stability fails on the bistable interval. Third, the coefficients of the extended system (9) are constrained only by the qualitative classical statements and not by measurement, and the representation of a chronic disorder by a sustained modification of self-production is one modelling choice among several, the alternative of constraining an organ to a fixed level not having been examined.

Several extensions suggest themselves. Computation of the first Lyapunov coefficient would determine the criticality of the Hopf bifurcation and hence whether the onset of oscillation is continuous. Two-parameter continuation of the stability boundary would locate the degenerate points at which the codimension-one bifurcations identified here interact. The intervention terms set to zero in [LH20] may be reinstated as control inputs, so that the model becomes a framework for the design of treatment: one then seeks the minimal intervention returning a diseased trajectory to the basin of attraction of the synchronous equilibrium, which corresponds to the classical principles of tonifying the mother and draining the child. Finally, the homeostatic target may be allowed to depend upon an organ's neighbours, through $\theta_i = \theta_i^0 + \alpha_i x_{\sigma(i)} - \beta_i x_{\rho(i)}$, rendering homeostasis contextual rather than constitutional; the present analysis applies to that variant without modification.

A Corrections to previously reported Lyapunov spectra

The symmetric bilinear model of [LH20] was reimplemented in the course of this work, and its equilibria, its Jacobian eigenvalues to each published decimal, and its reported period $T \approx 17.626$ were recovered. Three of the stability computations reported there do not, however, withstand examination, and the corrections are recorded here as they bear upon the stability of the bilinear synchronous equilibrium.

A structural constraint upon Lyapunov spectra

For a smooth autonomous system $\dot{x} = F(x)$, differentiation of the equation with respect to t gives $\frac{d}{dt}F(x(t)) = DF(x)F(x)$, so that $v(t) = F(x(t))$ is a solution of the variational equation. Its Lyapunov exponent is $\lim_{t\to\infty} t^{-1} \log \|F(x(t))\|$, which vanishes whenever the orbit is bounded, $\|F\|$ then remaining within a compact interval bounded away from zero. Every bounded trajectory that is not an equilibrium therefore possesses an exponent equal to zero. The argument does not apply at an equilibrium, where $F = 0$, which is why a stable equilibrium may possess only negative exponents. Both trajectories for which spectra are reported in [LH20] are bounded, one being a periodic orbit and the other confined by the invariants of Proposition A.1 below; both spectra must accordingly contain a zero exponent. Neither does:

$$\{0.3885, -0.1719, -0.3473, -0.6833, -0.9731\} \quad \text{and} \quad \{0.6700, 0.6606, -0.0761, -0.0794, -1.1751\}.$$

The periodic orbit is asymptotically stable

Recomputation at $a = b = c = d = g = h = 1$ by the variational method of Benettin et al. [Ben+80a; Ben+80b], with periodic QR reorthonormalisation and the initial transient discarded, yields $\{0.0011, -0.1232, -0.2721, -0.2728, -0.9998\}$: one zero exponent and four negative, which is the spectrum of an asymptotically stable limit cycle and not of the unstable periodic solution reported. The discrepancy is attributable to the singular-value method of [ZLZ05], employed in [LH20] following [Wol+85], having been applied over an interval including the transient, so that the rate of transient decay is measured in place of the asymptotic rate.

The equilibrium is neutrally stable

For the case in which only the restraining influences are retained, the reported exponents $\{0.0015, 0.0006, -1.315$ were interpreted as containing two positive values. Evaluation of the Jacobian at the attained final state $(0, 0, 0, 2.3426, 1.3038)$ yields instead $\{0, 0, -1.3038, -2.3426, -3.6464\}$, the two zeros being

exact, coordinates resting at $x_i = 0$ generating degenerate directions. The three negative eigenvalues agree with the reported values to within one percent, which identifies the reported positive entries as numerical error of order 10^{-3}. The correct conclusion is neutral stability, consistent with the observed convergence of the trajectory; as published, the case is described simultaneously as convergent and unstable.

The chaotic case is quasiperiodic

For $c = g = 1$ and $a = b = d = h = 0$ the bilinear model reduces to

$$\dot{x}_i = x_i(x_{i-1} - x_{i+1}), \qquad i \in \mathbb{Z}_5. \tag{10}$$

Proposition A.1. *System* (10) *admits the independent first integrals* $I_1 = \sum_i x_i$ *and* $I_2 = \prod_i x_i$.

Proof. Summation of (10) over i causes the right-hand side to telescope, whence $\frac{d}{dt}\sum_i x_i = 0$. On the positive orthant, $\frac{d}{dt}\log\prod_i x_i = \sum_i (x_{i-1} - x_{i+1}) = 0$. □

Both integrals were confirmed numerically, the relative drift being of order 10^{-15}. Two independent first integrals confine the motion to a three-dimensional invariant manifold, and the flow being volume-preserving, no attractor exists. The largest exponent computed over increasing intervals takes the values 0.0120, 0.0070 and 0.0038 at $T = 500$, 1000 and 2000, halving as T doubles, which is the behaviour of an exponent converging to zero rather than to the reported value 0.67. The motion is quasiperiodic upon a torus.

Consequence

The corrections do not affect the conclusion that the bilinear synchronous equilibrium is unstable, which follows from the eigenvalues $0.206 \pm 0.634i$ reproduced exactly. They establish that its attracting sets are limit cycles or boundary equilibria at which several organs vanish, neither of which represents sustained health.

B Numerical methods and validation

Solutions were computed by adaptive Runge–Kutta methods and, in stiff regions, by LSODA, with relative and absolute tolerances of 10^{-9} and 10^{-11}. Lyapunov spectra were computed by the variational method with periodic QR reorthonormalisation, an initial transient being discarded in each case. Bifurcation points were located by bisection upon the spectral abscissa and upon the discriminant of (4).

Three independent checks were applied. The Jacobians of both models, obtained analytically, were compared with finite-difference approximations at random non-synchronous states, the largest discrepancies being 2.4×10^{-7} and 3.2×10^{-7} respectively, consistent with truncation error. The closed-form spectrum (7) was compared with direct diagonalisation at 2582 randomly chosen parameter values, agreement being obtained to machine precision, and the stability classification derived from it agreed with that obtained numerically at all 2559 parameter values admitting a synchronous equilibrium. Finally, the prediction of equivariance that displacement of any organ yields an identical response was verified to within 10^{-14}, a discrepancy in which would indicate inconsistent implementation of the maps (1).

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